ASSIGNMENT-14.5

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Batch- 12

Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Lab Objectives

• Design functional, visually appealing web applications

Week7 -Friday

using HTML, CSS, and JavaScript with AI assistance.

• Apply responsive, accessible, and interactive design principles.

• Create practical UI components for real-world web applications.

• Use AI to optimize layout, UX, and performance

Task 1 – Image Gallery Website Create an image gallery web page.

Requirements:

• Display images in a grid layout.

• Use AI to:

o Optimize layout using CSS Grid.

o Add hover zoom or overlay effects.

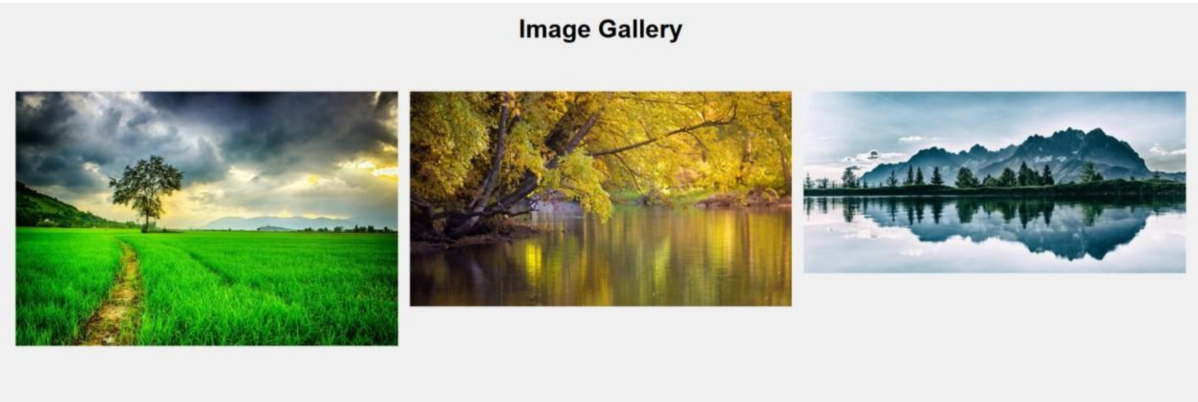
o Ensure mobile responsiveness.

```html
<!-- Create an image gallery web page with HTML and CSS.
 The gallery should display images in a grid layout. Optimize layout using CSS Grid.
o Add hover zoom or overlay effects.
o Ensure mobile responsiveness. give code the images block keep it empty i will paste the urls -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Image Gallery</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            background-color: #f0f0f0;
        }
        .gallery {
            display: grid;
            grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
            gap: 15px;
            padding: 20px;
        }
        .gallery-item {
            position: relative;
            overflow: hidden;
        }
        .gallery-item img {
            width: 100%;
            height: auto;
            transition: transform 0.3s ease;
        }
```

```html
        }
        .gallery-item:hover img {
            transform: scale(6.0);
        }
        .overlay {
            position: absolute;
            top: 0;
            left: 0;
            width: 100%;
            height: 100%;
            background-color: rgba(0, 0, 0, 0.5);
            color: white;
            display: flex;
            align-items: center;
            justify-content: center;
            opacity: 0;
            transition: opacity 0.3s ease;
        }
        .gallery-item:hover .overlay {
            opacity: 1;
        }
    </style>
</head>
<body>
    <h1 style="text-align: center; padding: 20px;">Image Gallery</h1>
    <div class="gallery">
        <!-- Add your image URLs here -->
        <div class="gallery-item">
            <img src="https://images.pexels.com/photos/236047/pexels-photo-236047.jpeg?cs=srgb&dl=clouds-cloudy-countryside-236047.jpg&fm
```

```html
</head>
<body>
    <h1 style="text-align: center; padding: 20px;">Image Gallery</h1>
    <div class="gallery">
        <!-- Add your image URLs here -->
        <div class="gallery-item">
            <img src="https://images.pexels.com/photos/236047/pexels-photo-236047.jpeg?cs=srgb&dl=clouds-cloudy-countryside-236047.jpg&f
            <div class="overlay">Image 1 Description</div>
        </div>
        <div class="gallery-item">
            <img src="https://www.pixelstalk.net/wp-content/uploads/2016/07/Download-Free-Pictures-3840x2160.jpg" alt="Image 2">
            <div class="overlay">Image 2 Description</div>
        </div>
        <div class="gallery-item">
            <img src="https://images.pexels.com/photos/459225/pexels-photo-459225.jpeg?cs=srgb&dl=daylight-environment-forest-459225.jpg
            <div class="overlay">Image 3 Description</div>
        </div>
        <!-- Add more gallery items as needed -->
    </div>

</body>
</html>
```

Output



Task 2 – Profile Card Generator

Create a profile card web page.

Requirements:

• Display user image, name, role, and social media links.

• Use AI to:

o Suggest modern UI colors and fonts.

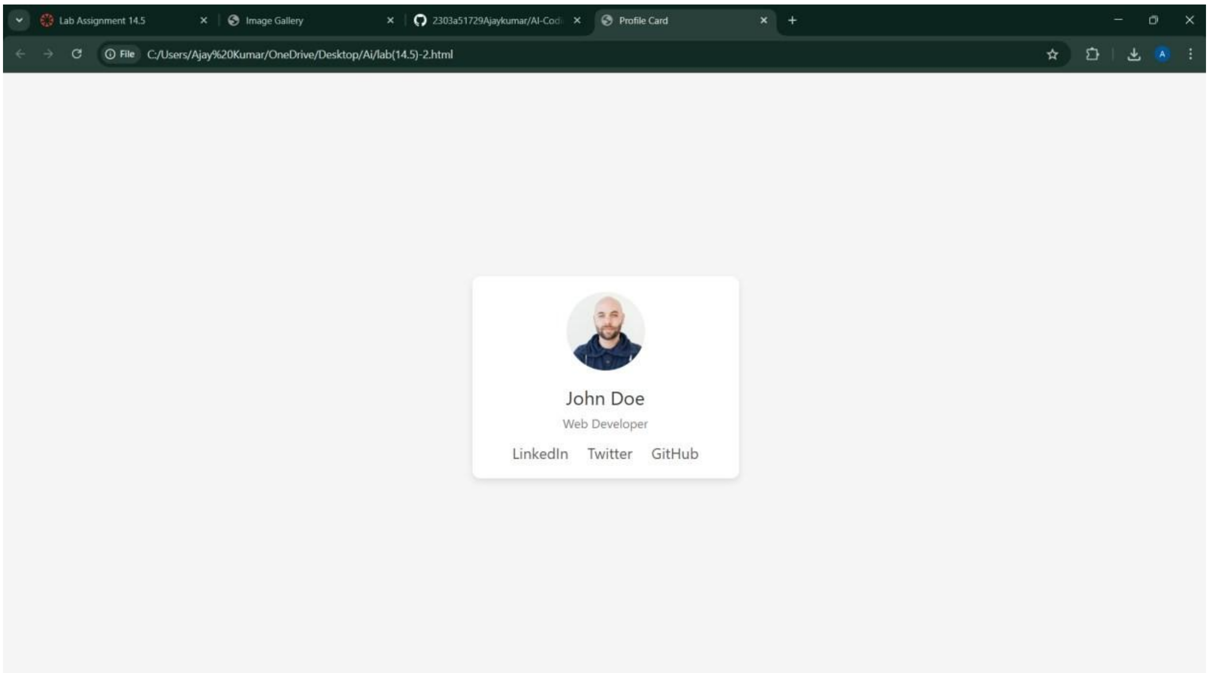
o Design the card using Flexbox.

o Add hover animation and shadow effects

```html
<!-- Task 2 – Profile Card Generator
Create a profile card web page.
Requirements:
• Display user image, name, role, and social media links.
• Use AI to:
o Suggest modern UI colors and fonts.
o Design the card using Flexbox.
o Add hover animation and shadow effects  give the code -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Profile Card</title>
    <style>
        body {
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
            background-color: #f5f5f5;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            margin: 0;
        }
        .profile-card {
            background-color: #ffffff;
            border-radius: 10px;
            box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
            width: 300px;
            text-align: center;
            padding: 20px;
            transition: transform 0.3s ease, box-shadow 0.3s ease;
```

```html
<html lang="en">
<head>
    <style>
        .profile-card {
        }
        .profile-card:hover {
            transform: translateY(-10px);
            box-shadow: 0 8px 16px rgba(0, 0, 0, 0.2);
        }
        .profile-image {
            width: 100px;
            height: 100px;
            border-radius: 50%;
            object-fit: cover;
            margin-bottom: 15px;
        }
        .profile-name {
            font-size: 1.5em;
            color: #333333;
            margin-bottom: 5px;
        }
        .profile-role {
            font-size: 1em;
            color: #777777;
            margin-bottom: 15px;
        }
        .social-links a {
            text-decoration: none;
            color: #555555;
            margin: 0 10px;
            font-size: 1.2em;
        }
        .social-links a:hover {
```

```html
<html lang="en">
<head>
    <style>
        .social-links a {
            font-size: 1.2em;
        }
        .social-links a:hover {
            color: #0073e6;
        }
    </style>
</head><body>
    <div class="profile-card">
        <img src="https://randomuser.me/api/portraits/men/75.jpg" alt="Profile Image" class="profile-image">
        <div class="profile-name">John Doe</div>
        <div class="profile-role">Web Developer</div>
        <div class="social-links">
            <a href="https://www.linkedin.com/in/johndoe" target="_blank">LinkedIn</a>
            <a href="https://twitter.com/johndoe" target="_blank">Twitter</a>
            <a href="https://github.com/johndoe" target="_blank">GitHub</a>
        </div>
    </div>
</body>
</html>
```



Task 3 -Develop a dynamic shopping cart system.

Requirements:

• Add, remove, and update product quantities.

• Display real-time total price.

• Use AI to:

o Manage cart state using JavaScript objects.

o Persist cart data in localStorage.

lab(14.5)-3.html > html > body > script > updateCart

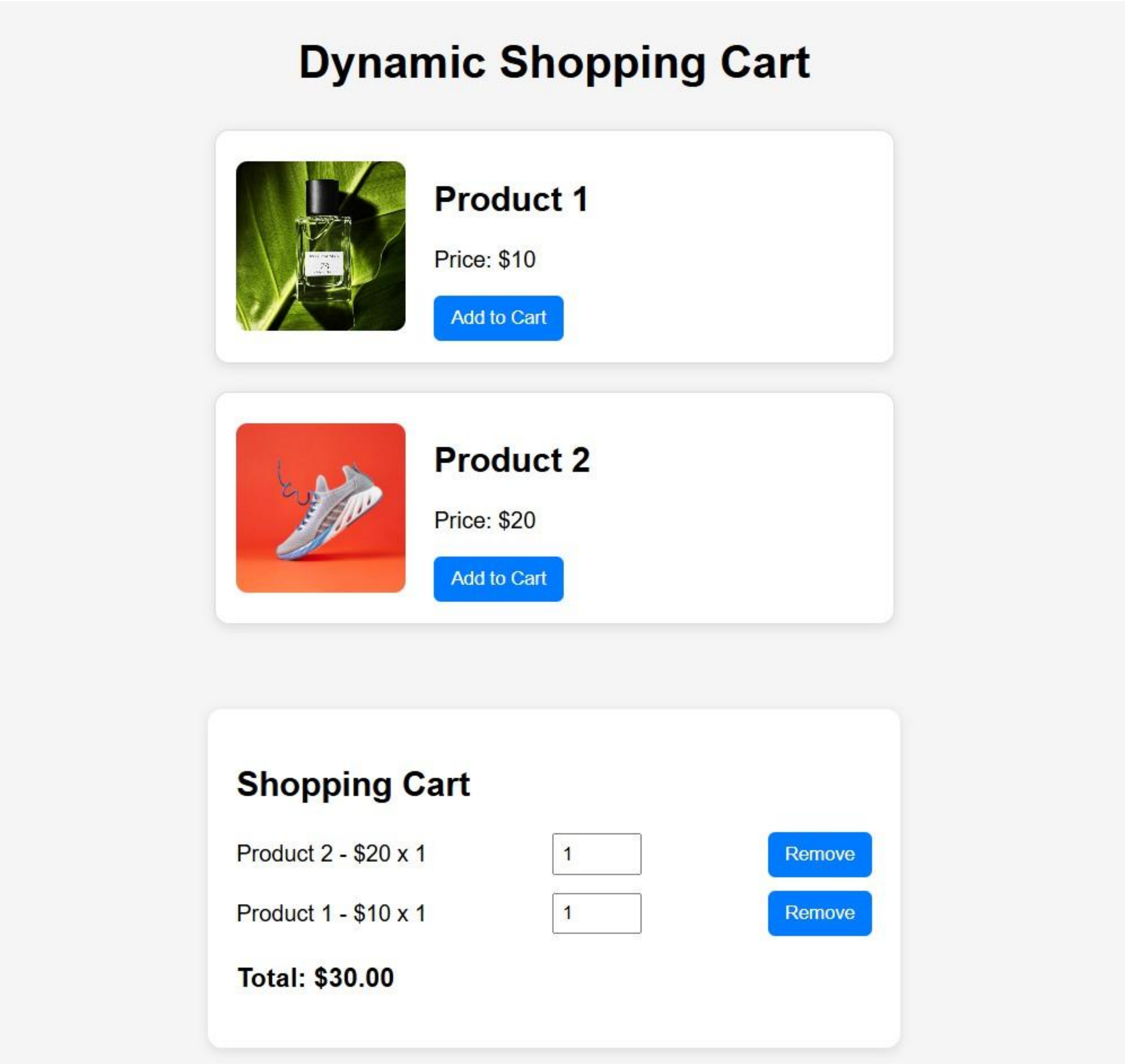
```html
<!-- Task 3 -Develop a dynamic shopping cart system.
Requirements:
• Add, remove, and update product quantities.
• Display real-time total price.
• Use AI to:
o Manage cart state using JavaScript objects.
o Persist cart data in localStorage. -->

<!-- i want images as well for the above prompt -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Dynamic Shopping Cart</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 20px;
            display: flex;
            flex-direction: column;
            align-items: center;
            background-color: #f5f5f5;
        }

        h1 {
            margin-bottom: 30px;
        }

        .product {
            display: flex;
```

```
    <style>
        }

        .product img {
            width: 120px;
            height: 120px;
            object-fit: cover;
            border-radius: 8px;
        }

        button {
            padding: 8px 12px;
            border: none;
            background-color: #007bff;
            color: white;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background-color: #0056b3;
        }

        .cart {
            margin-top: 40px;
            width: 450px;
            background: white;
            padding: 20px;
            border-radius: 10px;
            box-shadow: 0 2px 8px rgba(0,0,0,0.1);
```

```html
<body>

<h1>Dynamic Shopping Cart</h1>

<!-- Product 1 -->
<div class="product">
    <img src="https://i.pinimg.com/originals/35/94/79/3594793ee69d85ab8e82e780537fa83e.jpg" alt="Product 1">
    <div>
        <h2>Product 1</h2>
        <p>Price: $10</p>
        <button onclick="addToCart('Product 1', 10)">Add to Cart</button>
    </div>
</div>

<!-- Product 2 -->
<div class="product">
    <img src="https://d2v5dzhdg4zhx3.cloudfront.net/web-assets/images/storypages/primary/ProductShowcasesampleimages/JPEG/Product+Showcase
    <div>
        <h2>Product 2</h2>
        <p>Price: $20</p>
        <button onclick="addToCart('Product 2', 20)">Add to Cart</button>
    </div>
</div>

<!-- Cart Section -->
<div class="cart">
    <h2>Shopping Cart</h2>
    <div id="cart-items"></div>
    <h3>Total: $<span id="total-price">0</span></h3>
```

```html
<html lang="en">
<body>
<script>

    function updateCart() {
        const cartItemsContainer = document.getElementById('cart-items');
        cartItemsContainer.innerHTML = '';
        let totalPrice = 0;

        for (const product in cart) {
            const item = cart[product];
            totalPrice += item.price * item.quantity;

            const cartItem = document.createElement('div');
            cartItem.classList.add('cart-item');

            cartItem.innerHTML = `
                <span>${product} - $${item.price} x ${item.quantity}</span>
                <input type="number" value="${item.quantity}" min="0"
                       onchange="updateQuantity('${product}', this.value)">
                <button onclick="removeFromCart('${product}')">Remove</button>
            `;

            cartItemsContainer.appendChild(cartItem);
        }

        document.getElementById('total-price').textContent = totalPrice.toFixed(2);
        localStorage.setItem('cart', JSON.stringify(cart));
    }

    updateCart();
</script>
```

Output



Task 4 – Role-Based Access Control Application

Create a role-based web application.

Requirements:

• Admin, Editor, User roles.

• Conditional UI rendering.

• Use AI to:

o Design access control logic.

o Maintain secure state transitions.

o Create scalable UI components.

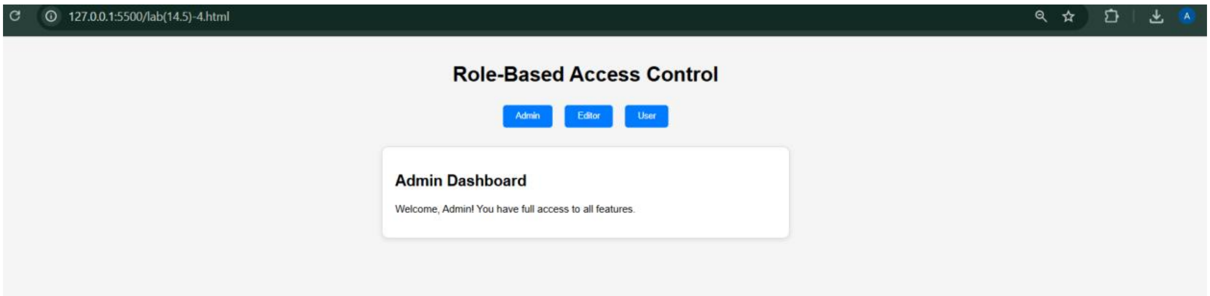
ab(14.5)-4.html > html > body > div.role-selection

```html
<!-- Task 4 – Role-Based Access Control Application
Create a role-based web application.
Requirements:
• Admin, Editor, User roles.
• Conditional UI rendering.
• Use AI to:
o Design access control logic.
o Maintain secure state transitions.
o Create scalable UI components. -->
<!-- give the responsive webpage for the above task give code -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Role-Based Access Control</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 20px;
            display: flex;
            flex-direction: column;
            align-items: center;
            background-color: #f5f5f5;
        }

        h1 {
            margin-bottom: 30px;
        }

        .role-selection {
```

```
<html lang="en">
<head>
    <style>
        .role-selection {
            margin-bottom: 30px;
        }

        .role-selection button {
            padding: 10px 20px;
            border: none;
            background-color: #007bff;
            color: white;
            border-radius: 5px;
            cursor: pointer;
        }

        .role-selection button:hover {
            background-color: #0056b3;
        }

        .content {
            width: 100%;
            max-width: 600px;
            padding: 20px;
            border: 1px solid #ddd;
            border-radius: 10px;
            background-color: white;
            box-shadow: 0 2px 8px rgba(0,0,0,0.1);
        }

        @media (max-width: 600px) {
            .content {
```

```
lab(14.5)-4.html > html > body > div.role-selection
<html lang="en">
<head>
    </style>
</head>
<body>
    <h1>Role-Based Access Control</h1>
    <div class="role-selection">
        <button onclick="setRole('Admin')">Admin</button>
        <button onclick="setRole('Editor')">Editor</button>
        <button onclick="setRole('User')">User</button>
    </div>
    <div class="content" id="content">
        Please select a role to see the content.
    </div>

    <script>
        function setRole(role) {
            const content = document.getElementById('content');
            if (role === 'Admin') {
                content.innerHTML = '<h2>Admin Dashboard</h2><p>Welcome, Admin! You have full access to all features.</p>';
            } else if (role === 'Editor') {
                content.innerHTML = '<h2>Editor Dashboard</h2><p>Welcome, Editor! You can edit content but have limited access to settin
            } else if (role === 'User') {
                content.innerHTML = '<h2>User Dashboard</h2><p>Welcome, User! You can view content but cannot make changes.</p>';
            }
        }
    </script>
</body>
</html>
```

Output:



Task 5 – Multi-Language Website

Design a multilingual website.

Requirements:

• Switch languages dynamically.

• Persist language preference.

• Use AI to:

o Structure translation files.

o Handle RTL/LTR layouts.

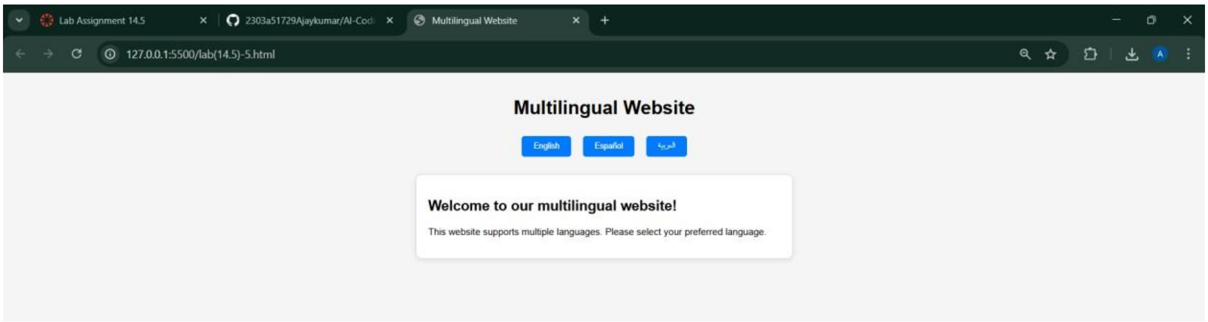
o Improve accessibility.

```
<!-- Task 5 – Multi-Language Website
Design a multilingual website.
Requirements:
• Switch languages dynamically.
• Persist language preference.
• Use AI to:
o Structure translation files.
o Handle RTL/LTR layouts.
o Improve accessibility. -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Multilingual Website</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 20px;
            display: flex;
            flex-direction: column;
            align-items: center;
            background-color: #f5f5f5;
        }

        h1 {
            margin-bottom: 30px;
        }

        .language-selection {
            display: flex;
```

```
        .language-selection button {
            padding: 10px 20px;
            border: none;
            background-color: #007bff;
            color: white;
            border-radius: 5px;
            cursor: pointer;
        }

        .language-selection button:hover {
            background-color: #0056b3;
        }

        .content {
            width: 100%;
            max-width: 600px;
            padding: 20px;
            border: 1px solid #ddd;
            border-radius: 10px;
            background-color: white;
            box-shadow: 0 2px 8px rgba(0,0,0,0.1);
        }
    </style>
</head>
<body>
    <h1>Multilingual Website</h1>
    <div class="language-selection">
        <button onclick="setLanguage('en')">English</button>
```

```
</head>
<body>
    <h1>Multilingual Website</h1>
    <div class="language-selection">
        <button onclick="setLanguage('en')">English</button>
        <button onclick="setLanguage('es')">Español</button>
        <button onclick="setLanguage('ar')">العربية</button>
    </div>
    <div class="content" id="content">
        <!-- Content will be dynamically updated based on language selection -->
    </div>

    <script>
        const translations = {
            en: {
                welcome: "Welcome to our multilingual website!",
                description: "This website supports multiple languages. Please select your preferred language."
            },
            es: {
                welcome: "¡Bienvenido a nuestro sitio web multilingüe!",
                description: "Este sitio web admite varios idiomas. Por favor, seleccione su idioma preferido."
            },
            ar: {
                welcome: "مرحبًا بكم في موقعنا متعدد اللغات!",
                description: "يدعم هذا الموقع عدة لغات. يرجى اختيار لغتك المفضلة."
            }
        };
```

Output:



Task 6 – Online Examination System

Create an online exam interface.

Requirements:

• Timer, questions, and submission.

• Prevent multiple submissions.

• Use AI to:

o Manage exam state.

o Improve exam UX.

o Ensure accessibility.

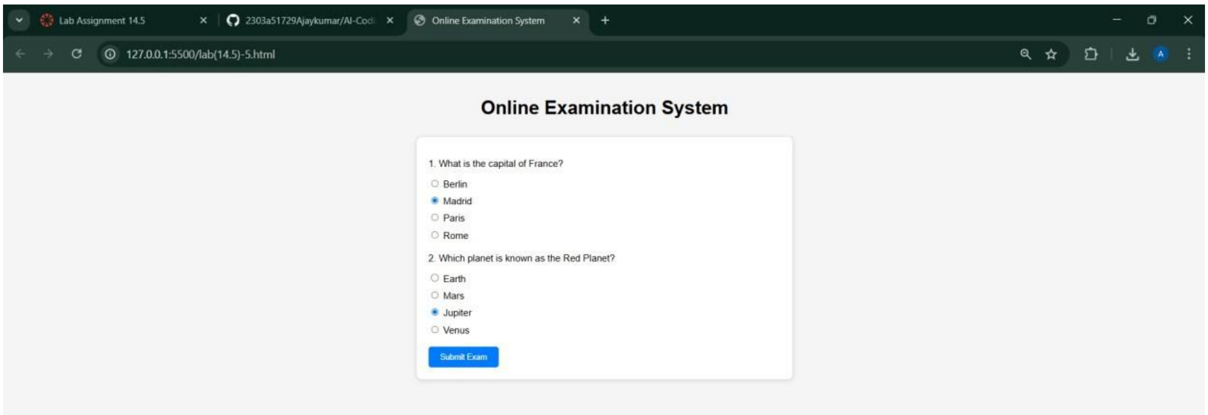
```html
<!-- Task 6 – Online Examination System
Create an online exam interface.
Requirements:
• Timer, questions, and submission.
• Prevent multiple submissions.
• Use AI to:
o Manage exam state.
o Improve exam UX.
o Ensure accessibility. -->
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Online Examination System</title>
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 20px;
            display: flex;
            flex-direction: column;
            align-items: center;
            background-color: #f5f5f5;
        }

        h1 {
            margin-bottom: 30px;
        }

        .exam-container {
            width: 100%;
```

```
    <style>
        }

        .options {
            display: flex;
            flex-direction: column;
            gap: 10px;
        }

        button {
            padding: 10px 20px;
            border: none;
            background-color: #007bff;
            color: white;
            border-radius: 5px;
            cursor: pointer;
        }

        button:hover {
            background-color: #0056b3;
        }
    </style>
</head>
<body>
    <h1>Online Examination System</h1>
    <div class="exam-container">
        <div class="question">
            <p>1. What is the capital of France?</p>
            <div class="options">
                <label><input type="radio" name="q1" value="a"> Berlin</label>
```

```
<html lang="en">
<head>
</head>
<body>
    <h1>Online Examination System</h1>
    <div class="exam-container">
        <div class="question">
            <p>1. What is the capital of France?</p>
            <div class="options">
                <label><input type="radio" name="q1" value="a"> Berlin</label>
                <label><input type="radio" name="q1" value="b"> Madrid</label>
                <label><input type="radio" name="q1" value="c"> Paris</label>
                <label><input type="radio" name="q1" value="d"> Rome</label>
            </div>
        </div>

        <div class="question">
            <p>2. Which planet is known as the Red Planet?</p>
            <div class="options">
                <label><input type="radio" name="q2" value="a"> Earth</label>
                <label><input type="radio" name="q2" value="b"> Mars</label>
                <label><input type="radio" name="q2" value="c"> Jupiter</label>
                <label><input type="radio" name="q2" value="d"> Venus</label>
            </div>
        </div>

        <button id="submit-btn">Submit Exam</button>
    </div>

    <script>
        const submitBtn = document.getElementById('submit-btn');
        let submitted = false;
```

```html
<body>

    <script>
        const submitBtn = document.getElementById('submit-btn');
        let submitted = false;

        submitBtn.addEventListener('click', () => {
            if (submitted) {
                alert('You have already submitted the exam.');
                return;
            }

            // Here you would typically gather the answers and send them to a server
            alert('Exam submitted successfully!');
            submitted = true;
        });
    </script>
</body>
</html>
```

Output:



Explanation: In this lab, we developed different web applications using HTML, CSS, and JavaScript with the help of AI. We created an image gallery, profile cards, shopping cart system, role-based access system, multilingual website, and an online exam system.
By using AI, we improved design, responsiveness, and user experience. We also learned how to make websites interactive, accessible, and efficient. This lab helped us understand how AI can make web development faster, smarter, and more effective.